

Experiment - 6

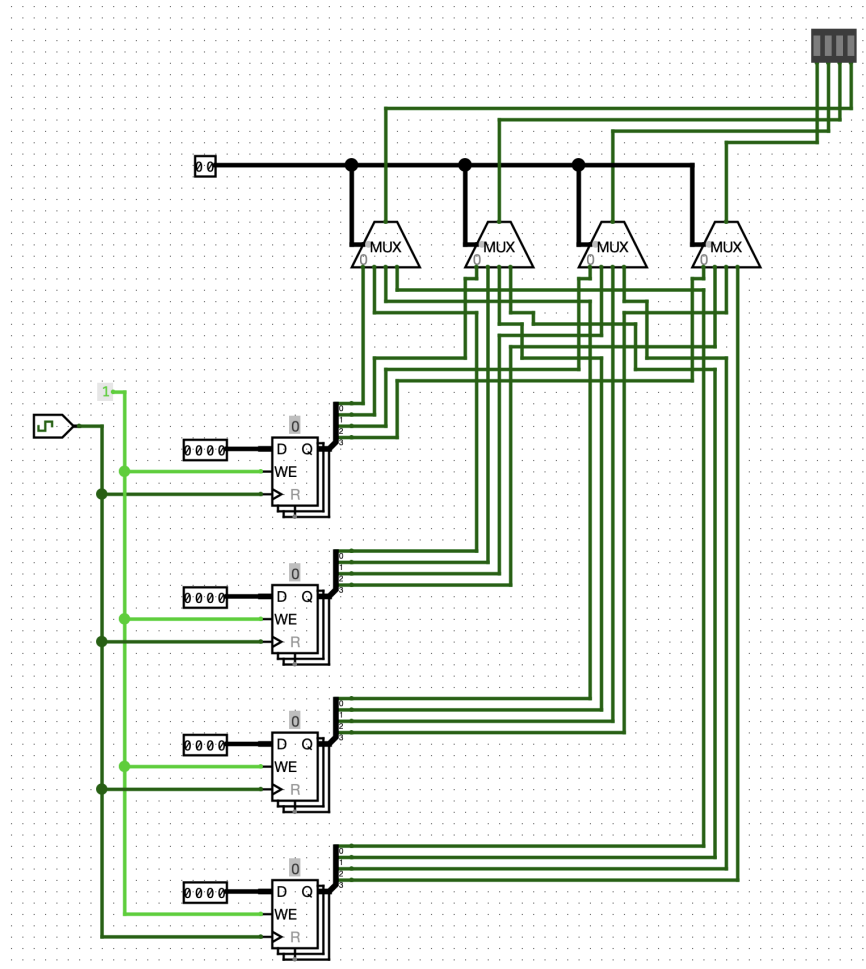
Aim

The objective of this experiment is to design and implement a 4-bit common bus system using multiplexers and registers in Logisim Evolution, and to verify correct data transfer between multiple registers through a shared bus controlled by selection logic.

Procedure

1. Opened Logisim Evolution and created a new project file for the experiment.
2. Placed four 4-bit registers on the canvas, which serve as the source and destination components for data transfer through the common bus.
3. Loaded distinct 4-bit binary values into each of the four registers to distinguish data during transfer verification.
4. Placed a 4-to-1 multiplexer for each bit position, resulting in four multiplexers to handle the full 4-bit wide bus.
5. Connected the corresponding output bit of each register to the respective input of each multiplexer, so that all four registers feed into the multiplexer bank.
6. Connected a common 2-bit select line to all four multiplexers simultaneously, ensuring that the same register is selected across all bit positions at once.
7. Wired the outputs of all four multiplexers together to form the 4-bit common bus line running across the canvas.
8. Connected the common bus output to the data input of all destination registers.
9. Added load/enable control signals to each register to determine which register latches the data present on the bus at any given time.
10. Applied a clock signal to the registers to synchronize the data loading operation.

11. Tested the circuit by setting the select lines to each of the four combinations (00, 01, 10, 11) in turn and verifying that the correct register's data appeared on the bus output.
12. Activated the load signal of a destination register while a source register was selected, and confirmed that the bus data was correctly latched into the destination register.
13. Repeated the transfer for all register combinations to fully verify the functionality of the common bus system.



4-bit Common Bus

Results

The simulation confirmed that data was correctly routed from each register onto the common bus based on the select line input, and destination registers successfully latched the bus data when their load signals were enabled. Only one register drove the bus at a time, consistent with the expected behavior of a common bus architecture.